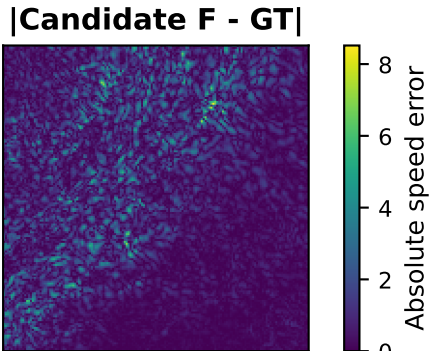
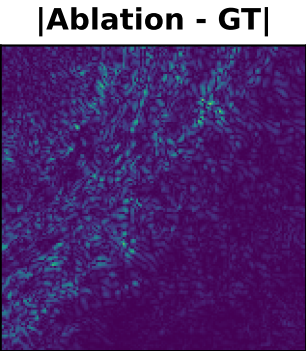
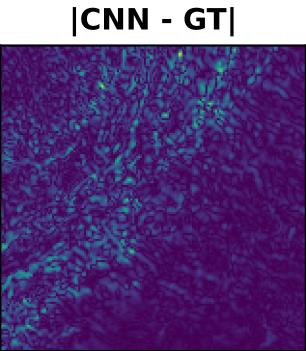
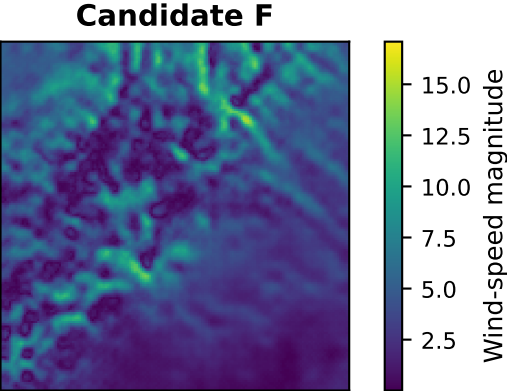
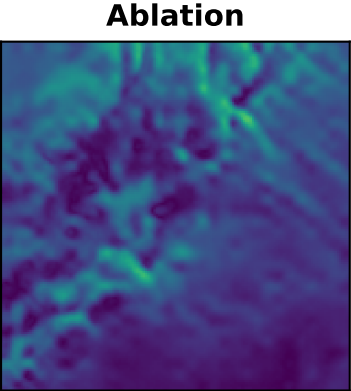
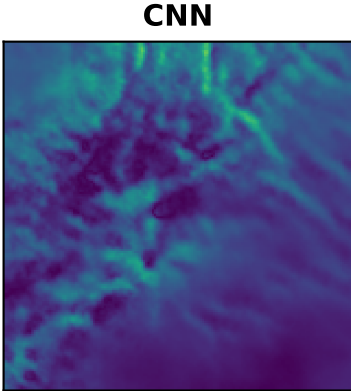
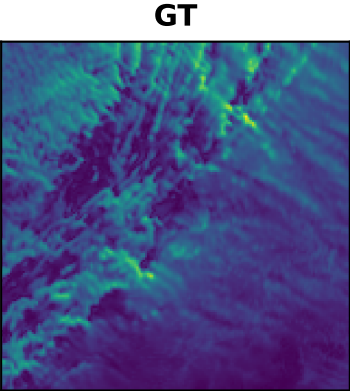
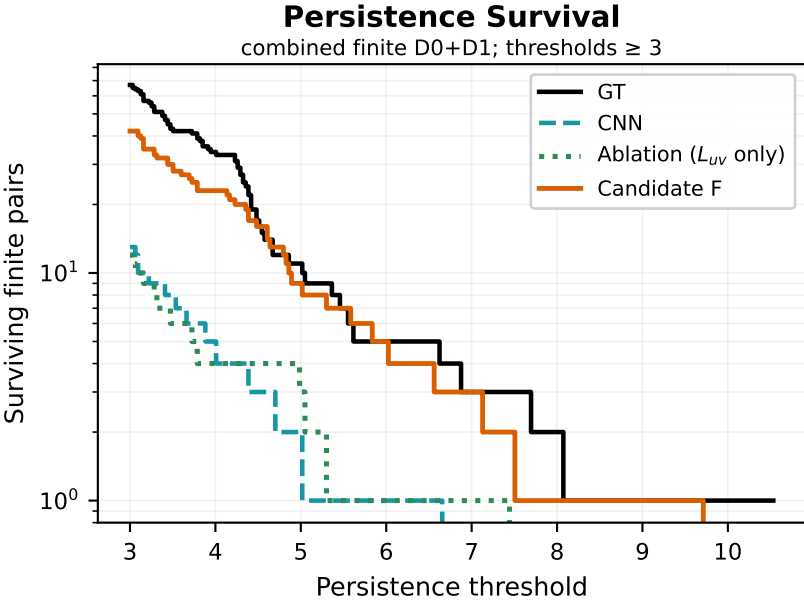
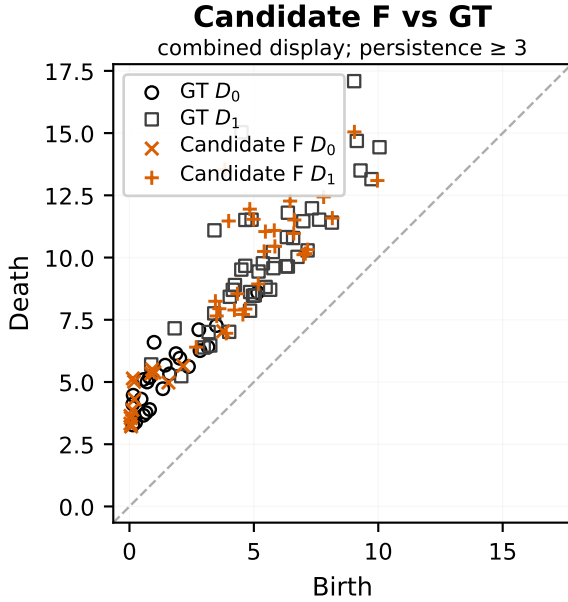
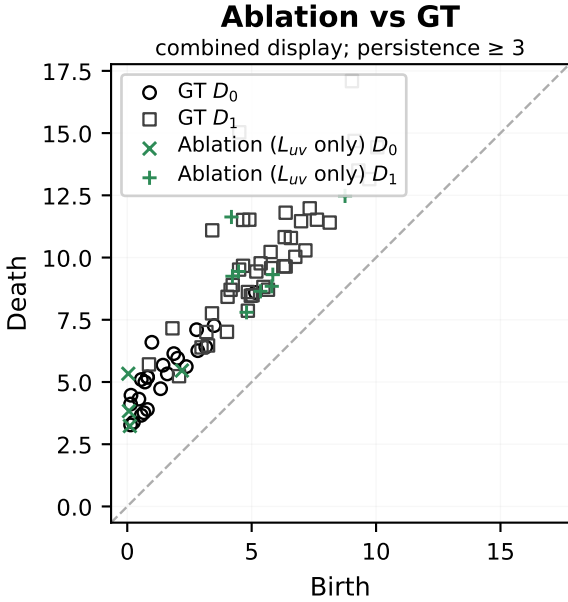
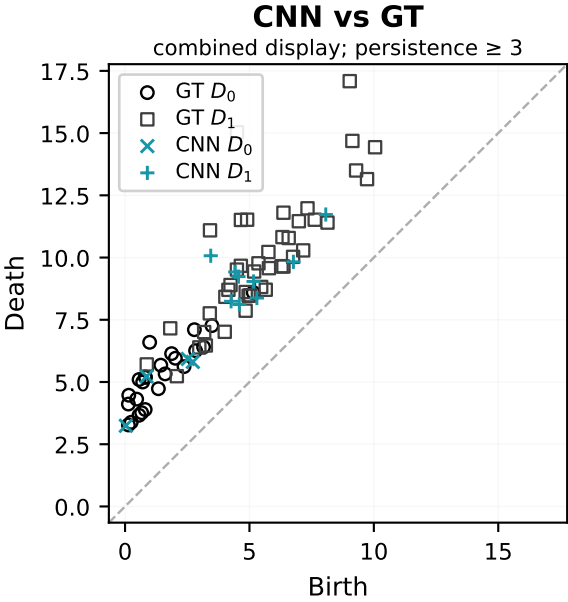


Sample 63: Topology-Aware Gains Beyond Matched Fine-Tuning

CNN = pretrained baseline; Ablation = matched L_{uv} -only fine-tuning; Candidate F = $L_{uv} + L_{grad}$ + repaired E2 supervision.



Topology-aware flow is better

Top 4 closeness ranks (lower is better)
 $W_{2,\infty}$: CNN 18.124 $\rightarrow$ Abl. 19.181 $\rightarrow$ F 13.145
 $W_{2,2}$: CNN 23.745 $\rightarrow$ Abl. 25.359 $\rightarrow$ F 16.741

Conventional fidelity gaps

F vs CNN (original closeness rank 1/168): $|\Delta\text{PSNR}|=0.463$ dB, MAE=8.4%, RMSE=3.3%
F vs L_{uv} ablation: $|\Delta\text{PSNR}|=1.238$ dB, MAE=16.3%, RMSE=14.8%

Matched-control interpretation

Candidate F is lower than the matched L_{uv} -only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.